[PZoom Deep Dive 9.3] Exhaust the contradiction where γ is a proper subset of both

Assume for contradiction that $\gamma \neq \alpha$ and $\gamma \neq \beta$.

Because $\gamma = \alpha \cap \beta$, we know that $\gamma \subseteq \alpha$ and $\gamma \subseteq \beta$. Since we are assuming γ is unequal to both, it must be a strict proper subset of both:

$$\gamma \subsetneq \alpha \quad \text{and} \quad \gamma \subsetneq \beta.$$

By Lemma 1 (Structure of Transitive Subsets), since γ is a transitive proper subset of the ordinal α, it must be an element of α. Similarly, since γ is a transitive proper subset of the ordinal β, it must be an element of β:

$$\gamma \in \alpha \quad \text{and} \quad \gamma \in \beta.$$

By definition of the intersection, an object belonging to both sets must belong to their intersection. Therefore:

$$\gamma \in (\alpha \cap \beta) \implies \gamma \in \gamma.$$

This states that γ is a member of itself. We can now construct the singleton set $K = \{\gamma\}$. Because $\gamma \in \gamma$, the set K has no $\in$-minimal element: its only member shares an element with K. This directly violates the Axiom of Regularity, which states that every nonempty set must contain an element disjoint from it under membership.

Thus, the assumption that γ is a proper subset of both ordinals is untenable. One of the sets must match γ entirely.

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